# 1. ATOMIC THEORY, FROM ANTIQUITY TO THE END OF THE NINETEENTH CENTURY

## I. MATTER AND ATOMS IN THE PHILOSOPHY OF ANTIQUITY

Nuclear physics is one of the most recently developed branches of physics. The term *nucleus* was first introduced by Rutherford about forty years ago, and the more detailed knowledge of the nuclei of atoms is only about fifteen years old. But the concept of the atomic structure of matter—the view that there exist certain smallest, ultimate, indivisible units, which are the basic building blocks of all matter—dates back to the philosophy of Antiquity, and was suggested by Greek philosophers as a daring hypothesis 2,500 years ago. Anybody who desires to understand something of modern atomic theory, will do well to study the history of the concept of the atom in order to become acquainted with the origins of those ideas which now have come to full fruition in modern physics. For this reason, the following lectures, the object of which is a description of the physics of the atomic nucleus, are prefaced by a short survey of the history of atomic theory.

The idea of the smallest, indivisible ultimate building blocks of all matter first came up in connection with the elaboration of the concepts of Matter, Being and Becoming, which characterized the first epoch of Greek philosophy. At the very dawn of ancient philosophy we find a remarkable statement by Thales, who lived in Miletus in the sixth century B.C.: He said that water was the source of all things. As Friedrich Nietzsche expounded, this sentence expresses three of the most essential and fundamental ideas of philosophy. Firstly, the question as to the source of all things; secondly, the demand that this question be answered in conformity with reason, without resort to myths or mysticism—in those times, no idea was regarded as more evident than that the source of all things must be sought in something material, such as water, and not in life—thirdly, the postulate that ultimately, it must be found possible to reduce everything to one principle. Thales' statement was the first expression of the idea of a fundamental substance, from which the whole universe had arisen, although in that age the word *substance* was certainly not interpreted in the purely material sense which we ascribe to it to-day.

In the philosophy of Anaximander, a pupil of Thales, who also lived and taught in Miletus, the idea of a fundamental polarity—the antithesis of Being and Becoming—was substituted for the concept of a single fundamental substance. Anaximander argued that if only one fundamental substance were